SW2

EventLens: Event Risk Research for Indices and Equities

by

Cheung Hau Yin Franklin Anthony Hou Liu Ngai Chiu Sze Wang Cheong

Advised by
Prof. Shuai Wang

Submitted in partial fulfillment
of the requirements for COMP 4981

in the

Department of Computer Science and Engineering
The Hong Kong University of Science and Technology 2026–2027

Date of submission: September 11, 2026

Abstract

Polymarket is an online prediction market where participants buy and sell YES or NO shares on real-world questions, such as whether the Federal Reserve will cut interest rates. A winning share pays one dollar, so its trading price can be interpreted as an approximate event probability. Orders are matched outside the blockchain, while completed trades settle through smart contracts on Polygon, a public blockchain.

This structure makes public Web3 data available beyond aggregate prices. Settlement records link trades to wallet addresses and outcome shares, allowing researchers to follow an address across events, assess its earlier decisions and trace observations to blockchain transactions. Public APIs and indexed blockchain datasets provide a practical route to collecting this evidence without operating a blockchain node. These data make it possible to test whether historically informative wallets contribute signals that market prices do not fully reflect, subject to sufficient historical coverage.

EventLens will investigate this question through a research dashboard for major US indices and selected equities, initially using Federal Reserve rate decisions. Users will compare market-implied probabilities with experimental wallet-informed estimates, inspect supporting activity, explore paper portfolio scenarios and request explanations from an AI assistant. The research will first evaluate event-probability accuracy, then test whether prediction-market signals improve forecasts of volatility, the size of future asset-price fluctuations. One linear regression baseline using recent squared returns will be compared with the same model augmented by prediction-market signals. Historical tests will use only information available at each forecast time. Public data make the analysis inspectable, while improved prediction remains a hypothesis to be tested.

Table of Contents

1 Introduction

1.1 Background and Motivation

Financial markets are repeatedly repriced around discrete events such as central-bank decisions, elections, and geopolitical developments. Investors can observe news, asset prices, and volatility indicators, but these sources do not directly show a continuously updated probability of the underlying event. This project aims to develop an machine-learning-assisted quantitative analysis framework EventLensthat extracts information from prediction markets and evaluates its value in forecasting financial-market volatility.

Prediction markets. Prediction markets such as Polymarket aggregate the beliefs of market participants into continuously updated probabilities. Market participants could trade YES and NO shares on the occurrence of events. A YES share pays one dollar if the specified outcome occurs and zero otherwise. A price of 60 cents can therefore be interpreted as an approximate market-implied probability of 60% [4, 5].

Wallets and blockchain records. Web 3 Prediction markets such as Polymarket records trade settlements on the blockchain. Its blockchain-based infrastructure provides transparent transaction histories that enable the behaviour and historical performance of individual participants (identified by their wallet addresses) to be analysed. This creates an opportunity to investigate whether information embedded in prediction-market activity, particularly from informed traders or "smart wallets", cancould provide useful signals for traditional financial markets.

1.2 Objectives

1. **Develop a smart-wallet identification model** using historical on-chain trading behavior to identify addresses whose decisions may add information beyond the market price.
2. **Construct smart-wallet-adjusted event probabilities** using wallet evidence and compare these experimental estimates with market-implied probabilities.
3. **Develop an event-to-asset relevance framework**: using statistical techniques and LLM research agent to identify traded events relevant to selected ETFs or indices.
4. **Evaluate whether Polymarket data provides incremental predictive valu** through historical backtesting and comparison against benchmark volatility forecasts, and present the results through an interactive AI-assisted dashboard

The primary target user is a research-oriented retail investor or finance professional who understands basic portfolio concepts but lacks the infrastructure to query blockchains, evaluate prediction-market participants, and construct event-conditioned risk models. EventLens is intended as a research and paper-decision tool rather than personalised financial advice.

Scope

This project will be separated in two phases. The first phase will focus on the minimum viable product (MVP), to validate the core concept, while the second phase will explore extensions and additional features. The scope of the project is summarized in Table 1, which outlines the minimum viable scope and the extended scopes.

Table 1: Minimum viable scope and extended scopes.

Dimension	MVP commitment	Extensions
Venue	Polymarket on Polygon	Cross-platform comparison, for example from Kalshi
Event coverage	Fed rates decisions	Other Macro and geopolitical events
Assets	Indices and ETFs	User-defined multi-asset portfolio
Horizons	1, 5 and 20 trading days	Additional term-structure horizons
Wallet model	Address-level, past-only scoring	Validated entity clustering

1.3 Literature Survey

Prediction markets and informative participants

Research on prediction markets suggests that prices may be disproportionately influenced by a relatively small group of informed or disciplined participants, suggesting that identifying these participants could provide valuable insights. Oliven and Rietz [9], for example, find that active market makers in the Iowa Electronic Markets behave more consistently with rational pricing conditions than typical price takers. This supports the “marginal trader” hypothesis: market accuracy may depend less on the average participant than on the traders willing and able to act against mispricing. Evidence from forecast aggregation is more mixed. Chen et al. [3] find that weighting forecasters according to past performance does not significantly outperform simpler pooling methods in their sports-prediction sample. By contrast, Atanasov et al. [1] show that performance weighting, temporal decay and recalibration can produce forecasts that outform standard prediction-market prices.

Recent Polymarket research provides more direct evidence of participant heterogeneity. Akey et al. [2] find that approximately 69% of users lose money and that the top 0.1% capture more than half of aggregate gains. Gómez-Cram et al. [8] classify approximately 3% of accounts as skilled. These accounts does not hold insider information but exhibit out-of-sample persistence and respond more rapidly to public information.

Prediction-market price as a proxy of event probability

Binary prediction-market contracts pay one unit if a specified outcome occurs and zero otherwise. Their prices are therefore commonly interpreted as event probabilities. Wolfers and Zitzewitz [4] argue that this interpretation can be a useful approximation under risk neutrality, limited risk aversion and sufficiently liquid trading. Nevertheless, the observed price remains an equilibrium outcome determined by traders' beliefs, wealth, preferences and the market mechanism, not automatically the physical probability that the event will occur.

Manski [14] demonstrates this limitation formally. With heterogeneous risk-neutral price-taking traders, the equilibrium price represents a budget-weighted quantile of the belief distribution rather than its mean. Consequently, the price may bound the average belief without identifying it exactly. Gao et al. [16] extend the analysis to heterogeneous risk-averse traders and utility-based market makers. Depending on the utility specification, the limiting price may resemble a geometric or power mean of beliefs and also depend on risk aversion, initial wealth and trading behaviour.

2 Methodology

2.1 Design

A guided research journey

A user selects an asset or hypothetical portfolio and chooses how far ahead to examine risk. EventLens presents relevant events, compares market probabilities with experimental wallet-informed estimates, and lets users inspect wallet activity before exploring scenarios. An AI assistant explains the evidence, and saved reports preserve the analysis.

For example, a user researching SPY over five trading days could inspect an upcoming rate decision, identify which tracked wallets are increasing YES or NO exposure, and compare the resulting risk scenarios. An incompatible announcement horizon or unsupported asset-response estimate will be explained before withholding the affected calculation.

Table 2: Core interface features and the decisions they support.

Screen or feature	What the user sees	What the user can do
Overview and watchlist	Supported assets, upcoming eligible events, source freshness and coverage.	Select holdings, choose a horizon and open an event.
Event detail	Question, outcome rules, announcement time, market probability and experimental estimate.	Compare estimates and inspect uncertainty and trading activity.
Smart-wallet tracker	Tracked addresses, prior event history, YES/NO exposure, recent activity and evidence quality.	Follow wallets, inspect their records and compare their contribution to an event signal.
Scenario studio	Gains and losses for each outcome, loss thresholds and average losses in the worst scenarios.	Change a stress assumption or compare paper allocations.
Research assistant	An explanation tied to the selected event, portfolio, horizon and evidence.	Ask follow-up questions and open the supporting sources.
Saved research	Analysis date, inputs, model and scenario versions, source links and limitations.	Revisit a frozen run and export a readable report with a machine-readable result.

Smart-wallet tracking and evidence inspection

Users can browse wallets in covered events, filter by history and activity, and follow public addresses in a research watchlist without connecting a wallet. Profiles show earlier resolved-event decisions, entry prices, subsequent price movements, reconstructed positions and an activity timeline. A provisional "smart" label denotes historically informative behaviour, supported by a dated score, sample size and coverage assessment.

Within an event, users can compare YES/NO holdings, agreement and whether a few addresses dominate activity. Contributions to the experimental estimate link to sources and blockchain transactions, distinguishing sampled verification from unchecked records. Personal watchlists do not change model-selection rules.

Results show their horizon, as-of time and missing evidence, separating observations from estimates and assumptions. Users can request explanations and save analyses. Insufficient wallet history leaves the adjustment unavailable while retaining supported market information.

Concept screens

Figures 1 and 2 illustrate the intended layout. They are conceptual interface images. Values are illustrative and do not represent research results.

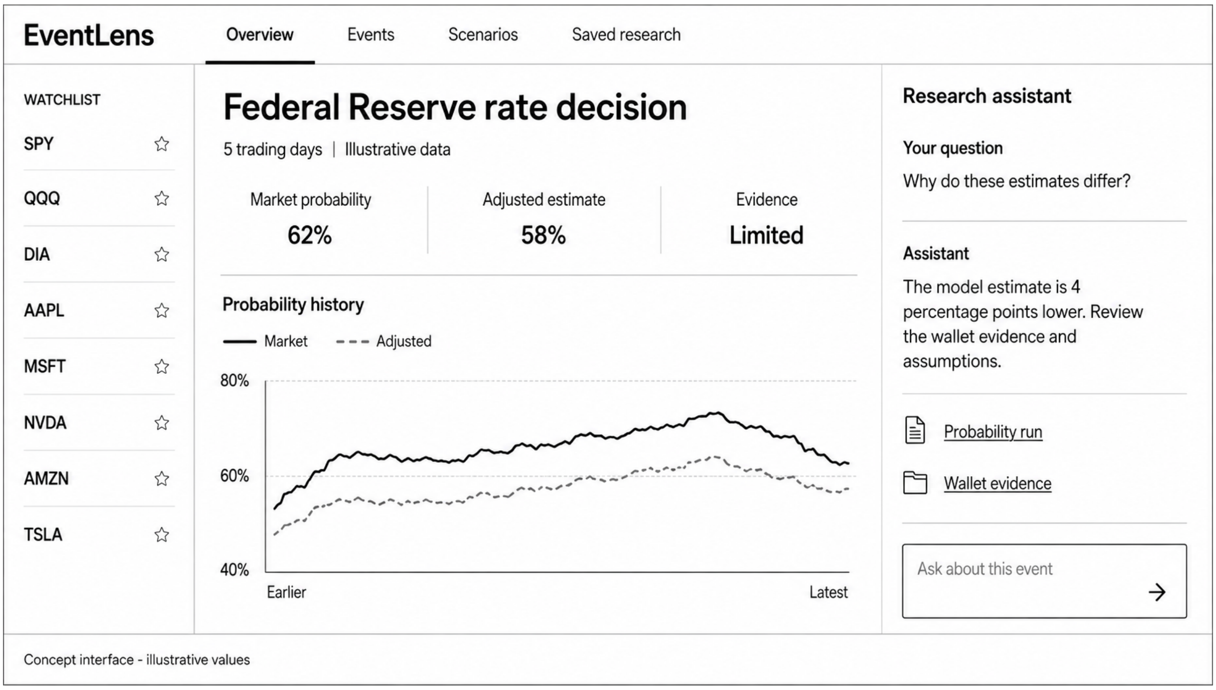

Figure 1: Overview concept: a watchlist, event probability comparison and contextual research assistant. All displayed values are illustrative.

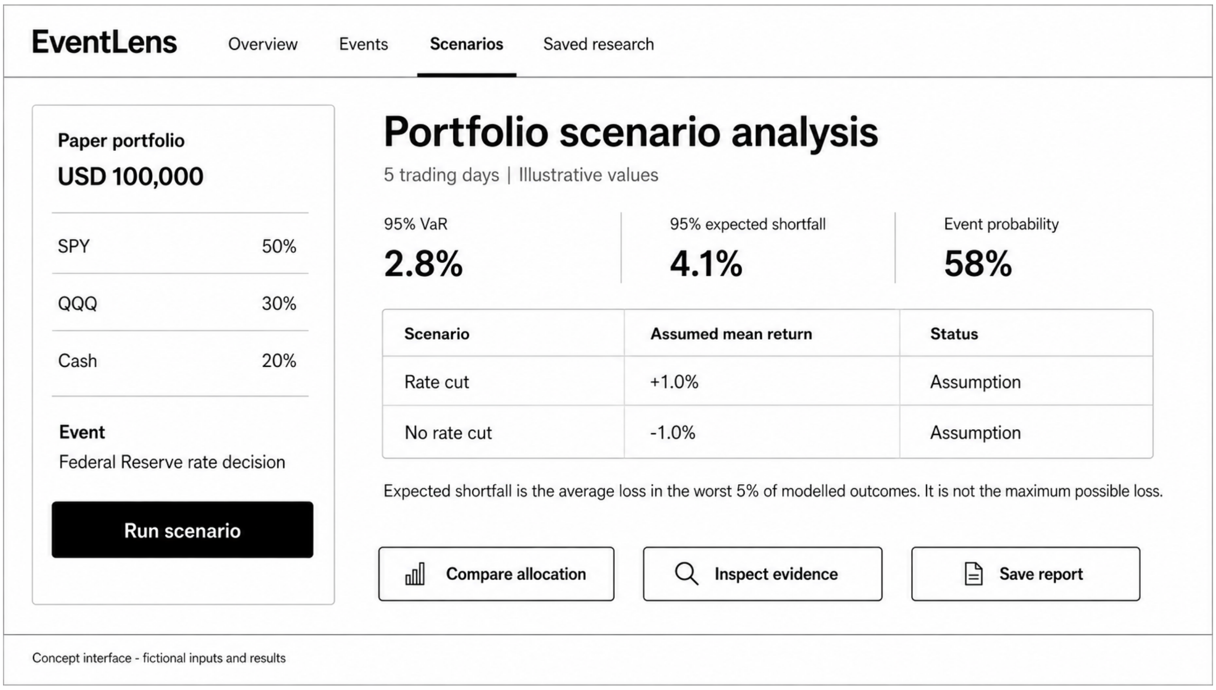

Figure 2: Scenario concept: portfolio inputs, clearly labelled model outputs and visible assumptions. The user compares paper scenarios without placing orders.

Three roles within one controlled workflow

The proposed assistant will use three logical roles: an **event research agent** to find relevant evidence, an **analysis coordinator** to request approved calculations, and an **explanation agent** to present results in context. These roles can share one model and orchestration service in the MVP. A separate autonomous agent infrastructure is unnecessary.

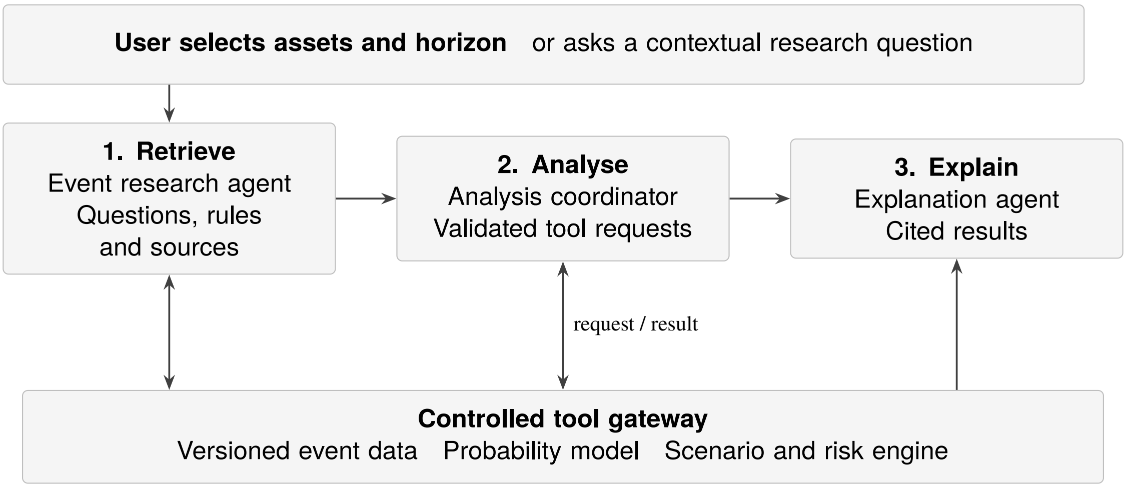

Figure 3: Proposed agent workflow. All quantitative results come from the same validated services used by the dashboard.

From a question to a supported answer

For "How could the next rate decision affect my SPY and MSFT exposure?", the assistant resolves the assets, horizon and analysis time. It retrieves an eligible event and proposes an event-to-asset explanation. Numerical responses must come from historical estimates or reviewed stress assumptions. Generated text alone is not sufficient evidence.

The coordinator can call tools such as `find_events`, `get_wallet_evidence`, `get_probability`, `run_scenario` and `compare_allocations`. The gateway checks asset coverage, weights, units, data freshness and horizon compatibility. It returns structured values with sources, model versions and a research-run identifier. The explanation then links its numerical claims to those results and separates observations, estimates and assumptions [12, 13].

Boundaries and review

Users confirm changes to their paper inputs and can inspect assumptions before running a comparison. Researchers review new event-to-asset mappings and choose model versions using validation results. The agent cannot alter wallet labels, tune against the final test period, invent missing prices, calculate risk in prose or submit live orders. Market descriptions are treated as untrusted data, including any embedded instructions.

When evidence is incomplete, the assistant will identify the gap and offer a supported view, such as the raw probability. Tool traces and saved reports make answers reviewable. Evaluation will cover citations, numerical agreement, abstention, user understanding, response time and cost.

Sources and their roles

The data pipeline will combine three kinds of evidence: prediction-market information, public blockchain settlement records and conventional financial data. These sources serve different purposes. Polymarket matches orders off-chain and settles trades on-chain, so an order-book snapshot is not itself a blockchain record [15].

Table 3: Planned sources, with access decisions made during the feasibility audit.

Source	Data to acquire	Use and limitation
Polymarket metadata API (Gamma)	Event definitions, rules, identifiers, outcome prices and available price history.	Define the forecast target and archive changes to mutable metadata [16].
Polymarket CLOB and activity Data API	Bid/ask snapshots, liquidity and public wallet activity.	Measure market quality and candidate wallet flow. Historical book depth may need prospective collection.
Goldsky Polymarket datasets	Indexed fills, matched trades and relevant position records.	Targeted historical reconstruction, subject to version coverage and access cost [17].
Polygon RPC	Transaction receipts, contract logs and block references.	Verify a stratified sample of settlements. Older state may require archival access.
Alpha Vantage or an approved equivalent	Daily equity/ETF prices, splits and dividends, with intraday data if available.	Proposed primary price-provider adapter. Adjusted daily data are a premium function. Coverage and academic-use terms must be checked [18].
Federal Reserve	Meeting dates, statements and announced rate decisions.	Establish announcement timing and validate event labels separately from platform resolution [19].

How on-chain data benefits EventLens

Public settlement records link activity to addresses and outcome tokens [15]. EventLens will use these histories to measure historically informative wallets' directional exposure, agreement and concentration across covered events. Separating transfers and token lifecycle operations from purchases avoids treating every balance increase as a new prediction.

Users can trace contributions to recorded transactions, and researchers can check indexed data against chain receipts. Addresses represent pseudonymous accounts, not verified people or independent decision-makers. Comparing models with and without wallet features will test whether this Web3 evidence improves forecasts. Provenance alone does not establish skill.

Turning source records into research inputs

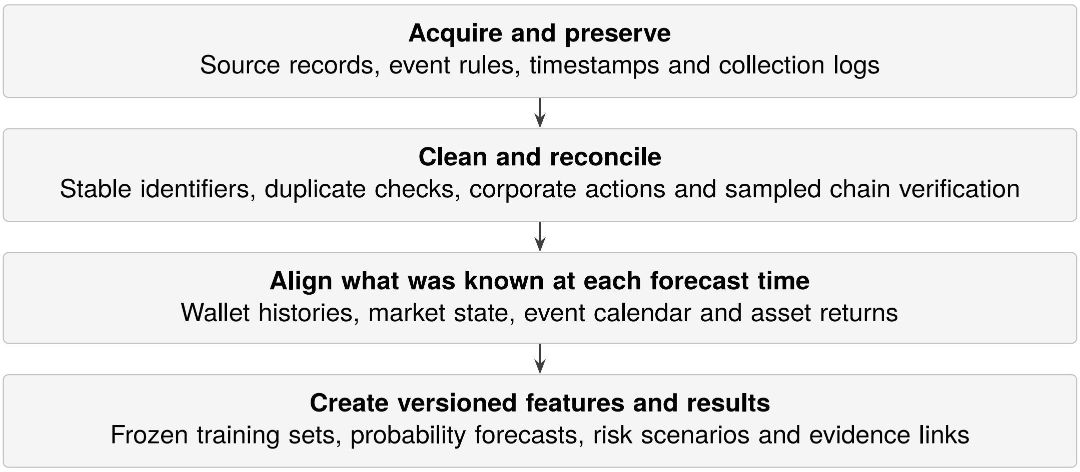

Figure 4: Processing pipeline. Raw evidence is retained so each transformation can be checked and repeated.

Preserve identifiers and time. Each event is linked to its markets, outcome tokens and settlement records. Raw records retain source time, collection time and source version. Announcement time, platform resolution time and data-availability time remain separate. Historical records downloaded later are admissible only when their content can be shown to have existed at the forecast cut-off. Mutable current rankings are not historical features.

Reconstruct activity consistently. Wallet analysis retains both participant perspectives, while market volume counts the economic trade once. A transaction may contain several logs, so transaction hash alone is not a unique record key. We will use chain, transaction, log and contract identifiers and verify sampled amounts against Polygon receipts. Position reconstruction must account for transfers, splits, merges and redemptions as well as trades. Incomplete positions receive a quality flag.

Align asset returns. Timestamps will be normalised to UTC while retaining the US trading calendar, daylight-saving changes and after-hours announcements. Splits and dividends will be handled consistently. Future adjustments or revised data must not change historical features without a recorded version. Missing bars will be flagged rather than filled with zero returns. Stale probabilities will not be carried across an announcement as if newly observed.

Make processing reproducible. Immutable source snapshots and checksums will accompany cleaned tables and Parquet research files. Collectors will use retries, rate limits and checkpoints. Replaying the same batch must not duplicate records. Each model run stores the data manifest, configuration, code version and random seed, and the interface will expose coverage gaps through its evidence panel.

Estimate the event probability

We will begin with a raw probability based on the YES bid/ask midpoint, recording spread, depth and timestamp. When both outcome books are usable, their midpoints can be normalised for consistency. A missing or one-sided book will be flagged, not silently replaced by an unrelated latest trade.

A simple regularised logistic model will calibrate this estimate using earlier resolved events. The wallet-informed model will then add a small number of signals: overall directional flow, the flow of historically informative wallets and market-quality controls. Coefficients will be fitted on training data, rather than chosen to produce a desired adjustment. Beta calibration is a possible robustness check if justified by the sample [14].

Select and track informative wallets

An informative wallet is an observed account whose earlier decisions contain information beyond the contemporaneous market price. Our first score will examine whether the wallet repeatedly chose the eventual outcome at informative prices, whether prices subsequently moved in its direction, and how broad its prior history was. Returns after fees will be a supporting diagnostic. Profitable liquidity provision alone is not proof of forecasting skill.

Repeated fills will be consolidated into wallet–event decisions. Balanced YES and NO holdings will be distinguished from directional exposure. Scores based on few independent events will be pulled towards the population average, a technique called shrinkage. A provisional eligibility rule is ten earlier resolved events across at least three announcement dates, with five- and twenty-event thresholds checked in sensitivity analysis. If the history cannot support this rule, the wallet signal will be unavailable rather than assigned artificial confidence.

Wallet eligibility and scores will use only records available and outcomes resolved by each forecast cut-off. New activity updates exposure. Newly resolved events update later scores. Frozen snapshots prevent today's successful wallets from being retrospectively selected for earlier tests.

Contributions combine earlier skill, direction, recency and capped size so one large wallet cannot dominate. Comparing total and eligible-wallet flow tests whether selection adds information. Profiles expose these components, timestamps, concentration and disagreement. New-wallet anomalies remain separate from historical skill.

Link events to assets

The event research agent may suggest that a rate decision matters to SPY or MSFT, but numerical impacts will come from matched historical event windows and conventional market controls. The mapping must account for surprise relative to the expectation already embedded in prices. Sparse asset-specific estimates will be pooled or replaced by explicitly reviewed stress ranges.

Wallet information may improve an event probability without improving an equity-risk forecast. The event-to-asset response is a separate estimate with its own uncertainty and evidence requirements.

Each mapping will retain its event definition, horizon, sample size, estimated response range and review status. A terminal YES/NO mixture will only be used when the economic announcement falls within the risk horizon. Otherwise the MVP will offer a compatible horizon or withhold that calculation. Correlated contracts from one meeting will not be treated as independent shocks.

Explore portfolio scenarios and tail risk

For a selected portfolio, the engine will combine the conditional return distributions for an event occurring (YES) and not occurring (NO). Let p be the selected event probability, μ a mean portfolio return and σ^2 its variance over the same horizon. The key calculations are:

$$\mu_{\text{mix}} = p\mu_{\text{YES}} + (1-p)\mu_{\text{NO}}, \tag{1}$$

$$\sigma^2_{\text{mix}} = p\sigma^2_{\text{YES}} + (1-p)\sigma^2_{\text{NO}} + p(1-p)(\mu_{\text{YES}} - \mu_{\text{NO}})^2. \tag{2}$$

The last term captures uncertainty about which outcome occurs. Consequently, a higher adverse-event probability can worsen expected loss without necessarily increasing variance. For multiple holdings, we will preserve cross-asset dependence when sampling returns, then apply the user's weights to each joint scenario.

The engine will report scenario returns, horizon volatility, a chosen loss-threshold probability, 95% VaR and 95% expected shortfall. VaR is a threshold, not the maximum loss. Expected shortfall will be calculated as the average of the worst 5% of the weighted loss distribution, including fractional threshold mass where needed. We will use empirical or heavy-tailed scenarios and check sensitivity to the probability and asset-response assumptions.

Paper comparisons will hold the event and model version constant while comparing the current allocation with a user-selected reduction in a supported holding and a corresponding increase in cash. Weights must sum to one and remain non-negative. Any displayed benefit must be considered alongside turnover, transaction-cost assumptions and estimation uncertainty.

Volatility forecasting and a trading extension

After estimating event probabilities, we will test whether prediction-market information improves a simple forecast of asset-price variation. The comparison uses one regression baseline and an augmented version of that same model. A separate trading extension would explore the forecasts' possible financial use.

Recurring event series and contract selection Polymarket serves as the sole source of prediction-market data for EventLens. The platform provides continuous tracking across recurring event families, including macroeconomic releases (e.g., FOMC rate decisions, CPI releases) as well as binary Polymarket contracts on equity index closing levels (e.g., S&P 500 / NASDAQ daily and weekly target closes) and precious metals price targets (e.g., Gold and Silver closes).

To maintain temporal consistency across continuous volatility forecasting horizons, an automated contract selection engine identifies and locks onto the closest-to-expiry market contract for any chosen event family. When event resolutions overlap or liquidity is fragmented across multiple contracts, the selection engine filters for active contracts meeting minimum volume and bid-ask spread thresholds, continuously rolling to the next active contract upon resolution.

One baseline and one augmented model The forecast target over H trading days will be the sum of squared daily log returns:

$$RV_{t,H} = \sum_{h=1}^{H} r_{t+h}^2. \tag{3}$$

This is a daily-data proxy for realised variance, not volatility itself. It measures the size of price movements without treating gains and losses as cancellations, but is noisy, especially for one day. Comparisons will use the same horizon and units. Any displayed volatility will be the square root of the corresponding variance estimate.

Baseline. We will fit a linear regression with an intercept and one input: the average squared daily return over the preceding 20 trading days. This provides a simple forecast based only on recent asset-price variation and requires no intraday data. A separate fit will be made for each supported asset and horizon using ordinary least squares on the training period.

Augmented model. The same regression will retain that input and add two event features: the market-implied probability p_{raw} and the wallet adjustment $p_{\text{adj}} - p_{\text{raw}}$. Here p_{adj} is the experimental wallet-informed probability. Both versions will use identical forecast dates, targets and training windows, restricted to dates with usable event evidence. The augmented model therefore tests the combined contribution of prediction-market information. The separate probability experiments assess whether wallet histories themselves add information.

Forecasts from both versions will use the same small positive floor, fixed before testing. Event features for training and evaluation will be generated using only data and probability-model fits available before each forecast date. Mean squared error (MSE) and quasi-likelihood loss (QLIKE) will compare forecasts on later, unseen observations [15]. These are evaluation measures, not claims of resistance to outliers. The interface will display the fixed comparison rather than let users change the baseline or feature set during evaluation.

Delta-hedged backtesting framework To demonstrate the practical utility of the forecasted realized volatility, EventLens runs an automated delta-hedged backtest. The backtest engine compares the model's forecasted realized volatility against current option-implied volatility (IV) from option market quotes to identify implied-versus-forecasted variance gaps.

When a variance gap is detected, the strategy simulates a delta-neutral position (e.g., ATM options hedged continuously against the underlying asset) to harvest the variance risk premium. Costs, bid/ask spreads, hedge frequency, and execution assumptions are included. This simplified backtest framework provides the user with an intuitive evaluation of forecast performance, displaying backtested Sharpe ratios, maximum drawdowns, and simulated trade logs directly on the research dashboard.

2.2 Implementation

We propose a modular Python backend and a TypeScript web frontend. One deployable application with a background worker will keep the four-person project manageable while separating collection, modelling, risk calculation and AI explanation.

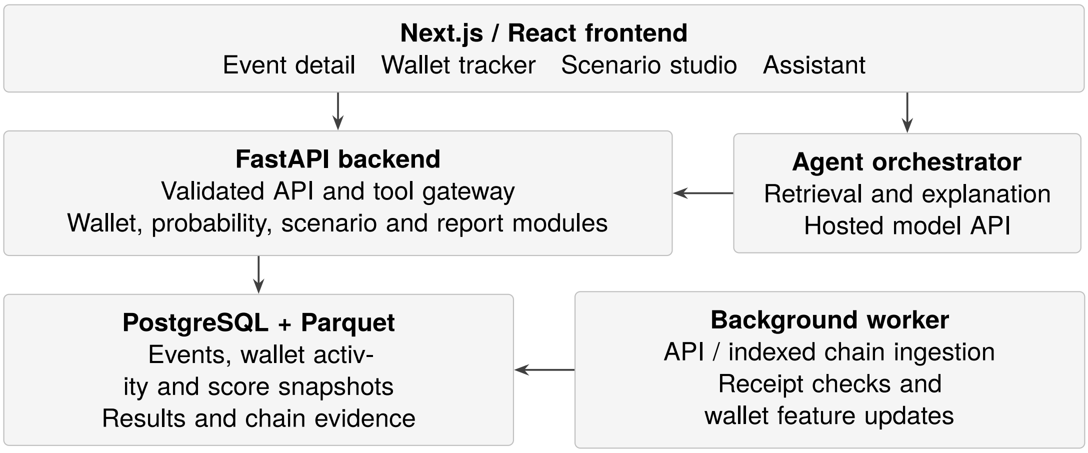

Figure 5: Proposed application architecture. The dashboard and assistant share the same calculation services and stored results.

Application services

Next.js and React will provide interactive views. FastAPI and Pydantic will validate requests and share calculation services with the assistant [21, 20]. One scikit-learn regression service will fit both volatility models. PostgreSQL will hold linked records and run metadata, while Parquet preserves raw evidence and research datasets. A Python worker and Docker Compose will support one manageable deployment.

Smart-wallet tracking pipeline

The worker will discover candidate addresses from covered markets and incrementally collect their relevant histories through activity APIs and version-compatible indexed chain data. It will map condition and outcome-token identifiers, deduplicate logs, and reconcile fills with transfers, splits, merges and redemptions. Checkpoints and retries make collection resumable. Polygon receipt checks will verify sampled records. Block references and verification status will accompany the evidence.

Separate tables will store wallet activity, position snapshots, dated skill scores, event-level contributions and user watchlists. Scores will be recomputed when eligible outcomes resolve. New activity will refresh exposure and aggregate flow. Each snapshot records its information cut-off, coverage and model version. Incomplete histories are flagged and excluded from calculations that require complete positions.

Backend endpoints will supply wallet search, profiles, watchlist updates, activity timelines and event contributions to both the tracker and assistant. The frontend will show the latest completed snapshot and refresh status. Watching an address does not make it model-eligible. Scoring follows the shared research rules.

Long analyses return a job identifier and status. Cached results include data, wallet-score, model, scenario, portfolio and horizon versions. Authenticated team access, server-side secrets, persistent volumes and backups protect the deployment. Hosted-model prompts and versions will be recorded, and direct research remains available when the assistant is unavailable.

2.3 Testing

Testing checks that the implementation is *correct*, that the code and pipeline do what they are meant to do – independently of whether the underlying research hypothesis turns out to be true.

Data and pipeline tests. Ingestion will be checked for schema validity, duplicate-free replay after an interrupted collection run, and reconciliation of a sample of collected records against source APIs and blockchain data. A point-in-time check will reject any feature or wallet history that uses information published after its forecast cut-off, since a lookahead leak is the failure most likely to silently invalidate every downstream result without producing an obvious error. Corporate-action handling (splits, dividends) and timestamp normalisation across time zones and daylight-saving changes will also be covered, since these are easy to get subtly wrong and hard to notice later.

Calculation tests. The probability model, wallet-scoring routine and the scenario/risk engine will be unit-tested against reference cases with known outputs: event probabilities of exactly zero and one, a missing asset, a one-sided or illiquid order book, and a singular or near-singular covariance matrix. API-level tests will enforce structural invariants – portfolio weights non-negative and summing to one, horizons restricted to supported values, VaR and expected shortfall computed consistently with their definitions – so that a broken invariant fails loudly rather than producing a plausible-looking wrong number.

Workflow and agent tests. End-to-end tests will exercise the full path from event selection to a saved, shareable report, including stale or unavailable upstream data. Agent-specific tests will cover malformed or out-of-scope tool requests and adversarial instructions embedded in retrieved market or news text, since retrieved content is treated as untrusted data rather than instructions. The standing requirement is that every number shown to the user, including any number referenced by the AI assistant, traces back to a saved, versioned result rather than to generated text. This will itself be checked by a small set of tests that compare assistant output against the underlying stored result.

Indicative metrics. Pass/fail against the reference cases and invariants above, automated-test coverage of the ingestion and calculation modules, zero duplicate records after a replayed or interrupted collection run, and successful, byte-identical reproduction of a prior run's results from its stored data manifest, configuration and random seed.

2.4 Evaluation

Evaluation asks whether the finished system meets the objectives in Section 1.2.

Table 4: How each objective will be evaluated (indicative and subject to refinement after the feasibility audit).

Objective	Question	Indicative metric
Wallet identification	Does the smart-wallet score capture information beyond chance?	Backtested accuracy/Brier score vs. a label-shuffled baseline, with sensitivity to the eligibility threshold
Adjusted probability	Does wallet-weighting improve on the raw market price?	Brier score and log loss across the baseline ladder (raw price → +liquidity → +anonymous flow → +wallet signal)
Volatility forecasting	Do event features improve the same regression model?	MSE and QLIKE for the recent-return baseline and its augmented version on identical forecast dates
Dashboard and assistant	Is the system usable, and does it stay evidence-linked?	Formative user study: task completion, numerical agreement between assistant and dashboard, correct abstention on insufficient data

Experimental design

Comparisons will use a chronological train/validation/test split, with each event's contracts and snapshots kept together and overlapping return windows purged at split boundaries, so that no test-period information leaks backward. The final test interval stays unopened until models and thresholds are fixed. Wallet scores, calibration and scenario inputs will use only information available as of each historical forecast time, matching the point-in-time discipline enforced in testing.

Given the likely small number of independent announcement dates, results will be reported with confidence intervals rather than as single numbers, and a broader macro sample would require a documented scope revision rather than an informal expansion mid-project. Multiple wallet-level significance tests will use a false-discovery correction rather than being read individually. A null result – no measurable improvement from wallet or event signals – is a valid and reportable outcome. Profitability is explicitly not a success criterion.

Usability evaluation

A small formative study (target 6–8 participants) will ask users to find an event, explain a probability difference, compare portfolio scenarios and identify a stated limitation, while we record task completion, time and errors. Fixed questions will compare three configurations – the full assistant, the dashboard alone, and a retrieval-only assistant without the calculation gateway – measured on source coverage, numerical agreement with the underlying stored result, correct abstention when evidence is insufficient, latency and estimated cost per query.

Risks to the evaluation itself

Table 5: Risks specific to producing a trustworthy evaluation, and planned responses.

Risk	Response
Too few independent events for reliable fitting	Report a descriptive pilot instead of a fitted comparison. Widen the event family only via a documented revision.
Selection bias in "smart" wallets	Freeze the wallet universe and score cut-offs before the test window opens. Apply shrinkage to low-sample scores.
Confounded asset response to an event	Use reviewed stress ranges and sensitivity analysis. Keep event-probability and event-to-asset uncertainty separate.
Multiple-comparison inflation	Apply false-discovery control across wallet-level and baseline-ladder tests before reporting significance.
Unsupported AI claims in the user study	Restrict tool arguments, link every numeric claim to a stored result, and test missing-data abstention explicitly.

Success requires reproducible data and experiment configuration, a controlled baseline comparison, a tested risk engine, and a usable, evidence-linked interface. Positive, negative and null findings will all be reported. Exact sample sizes, thresholds and the final participant count will be confirmed once the data-feasibility audit (Section 2.1) fixes what history is actually available.

3 Project Planning

3.1 Division of Work

Franklin, Anthony, Owen and Sunny will work across four tracks, each with a lead and a reviewer. Named ownership remains to be agreed. All members will contribute to literature review, reporting and presentations.

Table 6: Proposed work allocation, with named ownership to be recorded by the team.

Track	Responsibility	Deliverable
Data and blockchain	Source adapters, collection, reconciliation and quality reporting.	Versioned dataset and coverage report.
Probability research	Wallet scoring, calibration, baseline comparisons and leakage controls.	Reproducible probability experiments.
Scenarios and risk	Asset-response estimates, volatility, tail risk and reference tests.	Deterministic risk engine and application study.
Product and AI	Backend integration, frontend, agent tools and deployment.	Integrated dashboard and user evaluation.

Git will hold code, experiment configuration and separate LaTeX section files for collaborative editing. Generated files and secrets stay out of version control. Weekly reviews and an issue tracker will record owners, dependencies, meeting decisions and experiment changes.

3.2 GANTT Chart

The schedule is relative to project commencement and will be aligned with course dates. Data feasibility comes first. Interface work starts with sample data to identify integration problems early.

Table 7: Planning baseline. Shaded cells indicate the main work period.

Workstream	1–4	5–8	9–12	13–16	17–20	21–24
Scope, literature and feasibility						
Collection and reconciliation						
Wallet and probability models						
Event mapping and risk engine						
Interface and AI integration						
Evaluation and user study						
Report and reproducibility						

Milestones. Week 2: initial access and scope decision. Week 7: a versioned dataset and reconciliation report. Week 12: first raw-price, anonymous-flow and wallet-flow comparison. Week 18: working scenarios and risk calculations. Week 22: integrated evaluation and user tasks. Week 24: final report, demonstration and reproducibility package. Failed feasibility gates will narrow the model or coverage before optional work begins.

4 Hardware and Software Requirements

4.1 Hardware Requirements

The initial models and API-hosted language model do not require a local GPU. Development can be carried out on standard personal computers, with a modest shared server for scheduled collection and the demonstration.

Table 8: Proposed hardware provision.

Resource	Minimum provision	Preferred provision
Developer computer	Four-core CPU and 8 GB RAM	16 GB RAM for concurrent local services.
Storage	30 GB free disk space	Approximately 100 GB SSD space for bounded historical data.
Shared server	Local demonstration if necessary	2–4 vCPU and 8–16 GB RAM, subject to measured workload.
Network	Stable internet connection	Reliable access for collectors, hosted models and remote collaboration.

A self-hosted Polygon node is outside the initial scope. Hosted RPC and targeted indexed data will be used for collection and verification. Database and raw-data storage must persist across application updates and be backed up regularly.

4.2 Software Requirements

Table 9: Required software and external services.

Category	Software or service	Purpose
Runtime	Python and Node.js	Research services, collectors and web application.
Application	FastAPI, Pydantic, Next.js, React and TypeScript	Validated backend APIs and interactive interface.
Data and modelling	PostgreSQL, Parquet, Polars/Pandas, NumPy and scikit-learn	Storage, processing and initial statistical models.
External services	Polymarket APIs, Goldsky, Polygon RPC, equity data and hosted LLM	Event evidence, asset histories and research assistance.
Quality and deployment	Pytest, browser tests, Docker and Docker Compose	Automated verification and reproducible deployment.
Collaboration	Git, issue tracker and LaTeX	Source control, task allocation and shared documentation.

External costs may include adjusted equity histories, indexed blockchain access, archival RPC, a hosted language model and server hosting. The team will set a spending cap after checking sample data and provider terms during the feasibility stage. A paid historical options dataset is not required for the core project. Software dependencies, model versions and configuration will be recorded to support reproducibility.

5 References

[1] Atanasov, P., Rescober, P., Stone, E., Swift, S. A., Servan-Schreiber, E., Tetlock, P., Ungar, L., & Mellers, B. (2017). "Distilling the Wisdom of Crowds: Prediction Markets vs. Prediction Polls. Management Science, 63(3), 691–706. https://doi.org/10.1287/mnsc.2015.2374

[2] Akey, Pat and Gregoire, Vincent and Harvie, Nicolas and Martineau, Charles, "Who Wins and Who Loses In Prediction Markets? Evidence from Polymarket (March 18, 2026). http://dx.doi.org/10.2139/ssrn.6443103

[3] Chen, Y., Chu, C.-H., Mullen, T., & Pennock, D. M. (2005). "Information markets vs. opinion pools: an empirical comparison. EC '05 : *Proceedings of the 6th ACM Conference on Electronic Commerce* : Vancouver, Canada, June 5–8, 2005, 58–67. https://doi.org/10.1145/1064009.1064016

[4] J. Wolfers and E. Zitzewitz, "Prediction Markets," *Journal of Economic Perspectives*, 18(2), pp. 107–126, 2004. doi:10.1257/0895330041371321.

[5] J. Wolfers and E. Zitzewitz, "Interpreting Prediction Market Prices as Probabilities," NBER Working Paper 12200, 2006. doi:10.3386/w12200.

[6] G. W. Brier, "Verification of Forecasts Expressed in Terms of Probability," *Monthly Weather Review*, 78(1), pp. 1–3, 1950.

[7] T. Gneiting and A. E. Raftery, "Strictly Proper Scoring Rules, Prediction, and Estimation," *Journal of the American Statistical Association*, 102(477), pp. 359–378, 2007. doi:10.1198/016214506000001437.

[8] R. Gómez-Cram, Y. Guo, T. I. Jensen and H. Kung, "Prediction Market Accuracy: Crowd Wisdom or Informed Minority?" SSRN Working Paper 6617059, 2026. Working paper, not peer reviewed. doi:10.2139/ssrn.6617059.

[9] Oliven, K., & Rietz, T. A. (2004). "Suckers Are Born but Markets Are Made: Individual Rationality, Arbitrage, and Market Efficiency on an Electronic Futures Market." *Management Science*, 50(3), 336–351. doi:10.1287/mnsc.1040.0191.

[10] F. Victor, "Address Clustering Heuristics for Ethereum," *Financial Cryptography and Data Security*, pp. 617–633, 2020. doi:10.1007/978-3-030-51280-4_33.

[11] A. C. MacKinlay, "Event Studies in Economics and Finance," *Journal of Economic Literature*, 35(1), pp. 13–39, 1997.

[12] R. T. Rockafellar and S. Uryasev, "Optimization of Conditional Value-at-Risk," *Journal of Risk*, 2(3), pp. 21–41, 2000.

[13] T. G. Andersen, T.

[14] Manski, C. F. (2006). "Interpreting the predictions of prediction markets. Economics Letters, 91(3), 425–429. https://doi.org/10.1016/j.econlet.2006.01.004.

[15] A. J. Patton, "Volatility Forecast Comparison Using Imperfect Volatility Proxies," *Journal of Econometrics*, 160(1), pp. 246–256, 2011. doi:10.1016/j.jeconom.2010.03.034.

[16] Gao, J., Wang, Z., Wu, W., & Yu, D. (2025). "Price Interpretability of Prediction Markets: A Convergence Analysis. Operations Research, 73(1), 157–177. https://doi.org/10.1287/opre.2022.0417.

References (continued)

[12] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *Advances in Neural Information Processing Systems*, 2020. Proceedings and paper.

[13] S. Yao et al., "ReAct: Synergizing Reasoning and Acting in Language Models," *ICLR*, 2023. https://arxiv.org/abs/2210.03629.

[14] M. Kull, T. M. Silva Filho and P. Flach, "Beta Calibration: A Well-Founded and Easily Implemented Improvement on Logistic Calibration for Binary Classifiers," *AISTATS*, pp. 623–631, 2017. https://proceedings.mlr.press/v54/kull17a.html.

[15] Polymarket, "Order Lifecycle," documentation, accessed 5 September 2026. https://docs.polymarket.com/concepts/order-lifecycle.

[16] Polymarket, "Market Data Overview," documentation, accessed 5 September 2026. https://docs.polymarket.com/market-data/overview.

[17] Goldsky, "Indexing Polymarket with Goldsky," documentation, accessed 5 September 2026. https://docs.goldsky.com/chains/polymarket.

[18] Alpha Vantage, "API Documentation," including Daily Adjusted and corporate-action data, accessed 5 September 2026. https://www.alphavantage.co/documentation/.

[19] Board of Governors of the Federal Reserve System, "Meeting Calendars and Information," accessed 5 September 2026. https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm.

[20] FastAPI, "Features," documentation, accessed 5 September 2026. https://fastapi.tiangolo.com/features/.

[21] Next.js, "Server and Client Components," documentation, accessed 5 September 2026. https://nextjs.org/docs/app/getting-started/server-and-client-components.

6 Appendix A: Meeting Minutes

Initial project meeting

Date	8 August 2026
Location	Hong Kong
Attendees	Franklin, Anthony, Owen and Sunny
Purpose	Discuss the project direction, scope, evaluation approach and division of work.
Status	Preliminary discussion. Formal scope decisions and individual ownership remained open.

Discussion summary

The meeting considered a Web3 quantitative-trading and market-analysis agent. Subsequent research narrowed the proposed direction to Polymarket, public wallet activity, event-conditioned volatility and portfolio risk. The present proposal focuses the initial application on selected indices and popular US equities, with options paper trading retained as a possible extension.

Matters requiring confirmation

The team will confirm the initial asset and event universe, named workstream leads, data access and the schedule against official course milestones. Decisions and action items will be recorded with owners and deadlines at the next meeting.

Formal scope approval and named ownership were not recorded at this meeting.